JEWELLERY MANUFACTURING ERP

Workflow analysis & recommended software blueprint

Project vision
A connected operating system for jewellery manufacturers, retailers, production teams, karigars, inventory, accounts and dispatch.

This document combines the proposed business flow with the capabilities identified in the supplied reference video. It translates that flow into a practical software structure and implementation plan.

THE END-TO-END FLOW

Stage	What happens
01 Retailer B2B portal	Browse designs, select quantities and sizes, customize and submit an order.
02 CRM & order management	Create or match the retailer record, register the order and manage approval.
03 Production work order	Generate a linked work order with design, quantity, specifications and due date.
04 Gold issue & karigar chain	Issue material, assign stages and record every handoff between karigars.
05 Reconciliation & QC	Account for finished goods, returned metal, scrap, dust, process loss and differences; inspect quality.
06 Finished inventory	Receive approved pieces into stock, linked to the original work order and retailer order.
07 Dispatch & retailer tracking	Pack, invoice, ship and expose shipment progress and documents in the retailer portal.

1. What the reference workflow contributes

The reference workflow aligns with the original concept and reinforces that the product is more than an online ordering portal. It is a connected jewellery business platform with these core capabilities:

- **Digital B2B catalogue:** retailers browse designs, view specifications and place bulk orders.
- **Automatic CRM entry:** portal orders flow directly into the manufacturer's customer and order records.
- **Production and karigar chain:** work can be assigned by production stage and passed between a head karigar and sub-karigars.
- **Gold tracking:** record issued weight, returned weight, work-in-progress material, scrap and pending reconciliation.
- **Finished inventory:** approved jewellery is recorded against its originating order.
- **Retailer visibility:** customers can see production milestones, dispatch information and shipment tracking.

- **Sales communication:** enquiries, follow-ups and WhatsApp conversations can be organized alongside retailer records.

2. Recommended production and karigar model

A single product order may pass through multiple people and production stages. The system should support both direct manufacturer-to-karigar issue and a head-karigar model, depending on how the factory operates.

Role	Responsibility
Manufacturer / production manager	Creates work order and issues material to the responsible person.
Head karigar	Accepts responsibility for the job and may allocate sub-tasks or material to other karigars.
Stage karigars	Perform casting, filing, soldering, stone setting, polishing or other configured stages.
Final production / QC	Verifies quantity, weights, finish, purity and specifications; sends defects for rework.

Important: every material handoff must be a separate transaction. If 100 g is issued to a head karigar and 60 g is transferred to a casting karigar, the transfer must reduce the head karigar's custody and increase the casting karigar's custody. It must not be counted as a second 60 g of new gold issued by the manufacturer.

Suggested configurable production stages

Melting → Casting → Filing → Soldering → Setting → Polishing → Finishing → Quality Check → Final Weighing → Finished Goods Receipt

Each stage should capture assigned person, start and completion timestamps, input/output quantity, material movement, remarks, attachments and status. Stages should be configurable by product type.

3. Gold movement ledger – core differentiator

The gold ledger should be the backbone of the system. It must record custody and movement of metal from receipt into the factory through production and final dispatch. Track gross weight and fine-gold weight, with purity conversions where relevant.

Movement / measure	Example	System treatment
Gold issued	100.000 g	Issue voucher linked to work order and karigar
Transfer to another karigar	60.000 g	Internal custody transfer; not a new factory issue
Finished jewellery	95.000 g	Production output, with verified weight and purity
Raw gold returned	2.000 g	Return to available raw-metal stock
Scrap / dust recovered	1.500 g	Record separately with recovery status and assay if needed
Unreconciled difference	1.500 g	Exception requiring explanation and approval

Control rule: do not automatically label an unexplained difference as wastage. Require a reason code, supporting note and authorized approval. The reconciliation model must account for material still in work-in-progress and avoid double-counting transfers.

Recommended gold ledger records

- Opening stock, purchases, supplier, purity, gross weight and fine-gold weight.
- Issue vouchers: karigar, work order, issued quantity, date, issuing user and acknowledgement.

- Return vouchers: finished output, raw return, scrap, dust, rejected or semi-finished pieces.
- Internal transfer vouchers between head karigars, sub-karigars and production stages.
- Reconciliation: issued material versus output, returns, recoveries, approved process loss and unresolved variance.
- Immutable transaction history with user, timestamp, reference document and reversal/correction trail.

4. Finished goods, quality and dispatch

When a work order is completed, the system should not immediately make the goods available for sale. The goods should pass through quality approval and a controlled finished-goods receipt.

Finished goods record

- Unique inventory ID, design/SKU, product category, order and work-order references.
- Quantity, size, gross weight, net gold weight, purity, stone weight and stone details.
- Making charges, manufacturing date, storage location and quality approval.
- Movement history: receipt, reservation for order, packing, dispatch, return or adjustment.

Quality control checklist

- Verify design and customization against the retailer's order.
- Check quantity, dimensions, size, weight and purity as applicable.
- Inspect finishing, polish, soldering and stone setting.
- Record pass, reject or rework decision, defect reason, inspector and timestamp.

Dispatch process

Reserve approved stock → create packing list → record final weights → generate invoice and delivery challan → enter courier and tracking details → mark dispatched → sync shipment events → confirm delivery.

5. Software modules and dashboards

Module	Primary responsibility
B2B retailer portal	Catalogue, retailer-specific pricing, bulk orders, customization, order history and shipment visibility.
CRM & sales	Retailer profiles, enquiries, communication, follow-ups, sales ownership, credit and purchase history.
Order management	Order approval, commercial terms, expected delivery, amendments, cancellation and status history.
Production ERP	Work orders, stage routing, assignments, capacity, due dates, delays, rework and completion.
Karigar management	Karigar profiles, hierarchy, work allocation, material custody, output and making-charge records.
Gold & inventory	Raw metal, work-in-progress, scrap, recovered gold, finished goods and transaction ledger.
Quality control	Inspection checklists, approvals, rejection, rework and traceability.
Accounts	Invoices, receipts, advances, GST, retailer balances, supplier and karigar payments.
Dispatch	Packing, shipment documents, courier tracking, delivery confirmation and returns.
Reports & analytics	Order pipeline, production performance, gold reconciliation, stock, sales and receivables.

Manufacturer admin navigation

Dashboard · CRM & Retailers · B2B Orders · Work Orders · Production · Karigar Management · Gold Movement Ledger · Raw & Finished Inventory · Quality Control · Dispatch & Shipping · Accounts & Payments · WhatsApp & Sales · Reports & Analytics · Settings & Permissions

Retailer portal navigation

Design Catalogue · New Order · My Orders · Production Tracking · Shipment Tracking · Invoices · Payments & Outstanding Balance · Enquiries & Support

6. Roles and access control

Role	Typical access
Super admin / owner	Business-wide configuration, dashboards, approvals and reports.
Sales manager	Retailers, enquiries, quotations, CRM and orders.
Order manager	Order validation, approval and work-order creation.
Production manager	Production routing, stage assignment, karigar allocation and progress.
Inventory manager	Gold receipts, issues, returns, transfers and stock control.
Accountant	Invoices, receipts, GST, balances and financial reports.
Quality manager	Inspection, rejection, rework and release approval.
Dispatch manager	Packing, documents, courier and shipment updates.
Karigar	Only assigned jobs, instructions, acknowledgements and permitted progress updates.
Retailer	Only their own catalogue access, orders, invoices, queries and shipments.

Access must be permission-based and scoped to the relevant manufacturer, department, work order or retailer. Retailers must never see internal gold balances, karigar payments or another retailer's information.

7. Important business rules

1. Every retailer order has a unique order ID and an auditable status history.

2. Every approved production order links to one or more work orders.

3. Every gold issue, return and transfer is linked to a person, work order and transaction voucher.

4. A work order cannot be closed until required material reconciliation is complete or an authorized exception is approved.

5. Finished goods enter available inventory only after quality approval.

6. Dispatch reduces or reserves finished inventory and records shipment references.

7. Corrections use controlled reversals or adjustment entries; never silently delete stock movements.

8. Support multiple purities and record both gross and fine-gold weights where relevant.

9. Use precise decimal storage for weights (at least three decimal places; choose precision after operational validation).

10. Keep metal accounting separate from money accounting. A retailer's financial balance is not the same as a metal balance.

11. Maintain audit logs for sensitive actions, approvals, stock changes and financial transactions.

8. Recommended technical architecture

A web-based, API-driven application is a practical starting point. Keep the architecture modular so the manufacturer admin, retailer portal and future mobile experiences use the same business logic.

Layer	Recommendation
Frontend	React + Vite + Tailwind CSS; responsive admin and retailer interfaces.
Backend	Laravel REST API with modular domain services.
Database	MySQL with transaction-safe stock and ledger operations.
Authentication	Secure authentication, role-based permissions and tenant scoping.
Documents	Secure storage for product images, vouchers, invoices, QC photos and shipment documents.
Notifications	In-app and email first; WhatsApp Business API integration where approved and available.
Deployment	Cloud VPS with backups, monitoring, access controls and documented recovery procedures.

Build SaaS-ready, even if the first deployment is for one manufacturer: every business record should be scoped to a manufacturer/tenant, with strict data isolation and tenant-aware permissions.

9. Phased implementation plan

Phase	Scope	Outcome
1	Authentication, roles, retailer CRM and B2B catalogue	Retailers can log in, browse and submit orders.

Phase	Scope	Outcome
2	Order workflow, approvals and work-order generation	Orders move from sales to production without re-entry.
3	Gold ledger, karigar hierarchy and material transfers	Every gram has a recorded custody trail.
4	Production stages and finished inventory	Work-in-progress and completed goods are traceable.
5	Quality control, accounts, invoices and dispatch	Orders can be checked, billed and shipped.
6	Retailer tracking portal and notifications	Retailers can self-serve order and shipment updates.
7	WhatsApp, advanced analytics and AI assistance	Improve communication and operational insights.

10. Decisions to confirm before database design

These operational choices affect the ledger, production routing and portal scope. Confirm them with the manufacturer before implementation.

Decision	Options to validate
Karigar model	Head karigar manages sub-karigars; direct issue to individual karigars; or both.
Gold accounting	Gross weight only; gross + purity + fine-gold weight; or those measures plus separate scrap, dust and process-loss accounting.
Retailer access	Responsive web portal first; mobile app; or web first with app planned later.
Production routing	Fixed stages by product type or configurable routing per work order.
Metal settlement	How purity, solder/alloy additions, stones, dust, scrap, recoveries and allowable process loss are measured and approved.
Commercial model	Retailer-specific rates, making charges, advances, credit limits and payment terms.

11. Final product definition

Jewellery Manufacturing Operating System — a centralized platform that connects retailer ordering, CRM, production planning, karigar workflows, gold accountability, inventory, quality control, accounting and dispatch.

The defining differentiator should be end-to-end traceability: from the retailer's selected design and original order, through every work order and material handoff, to the finished jewellery item and final shipment.

Prepared as a workflow and software planning document. Examples and weights are illustrative; actual metal accounting rules should be validated with the manufacturer's production and accounts teams before implementation.